

Deep Learning Project

PIGG: Personalized Interactive GUI Grounding

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CSE403: Deep Learning
Fall Semester, 2025, UNIST

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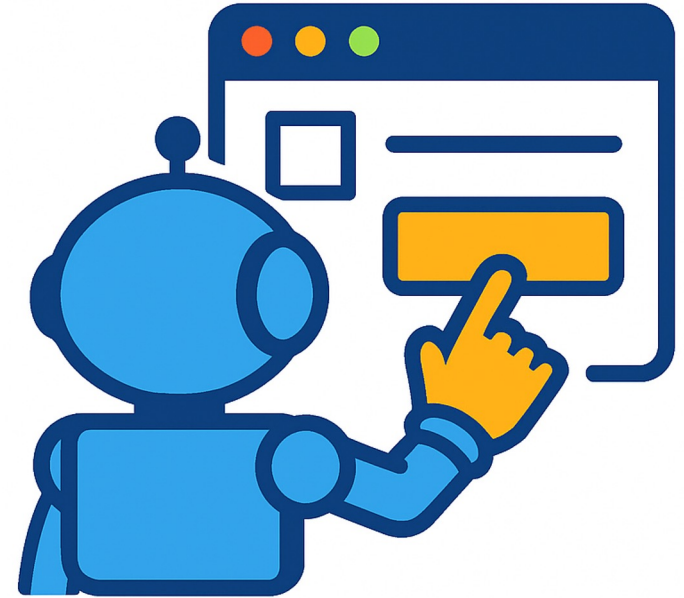
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Problem Definition

What is GUI Grounding?

Problem Definition

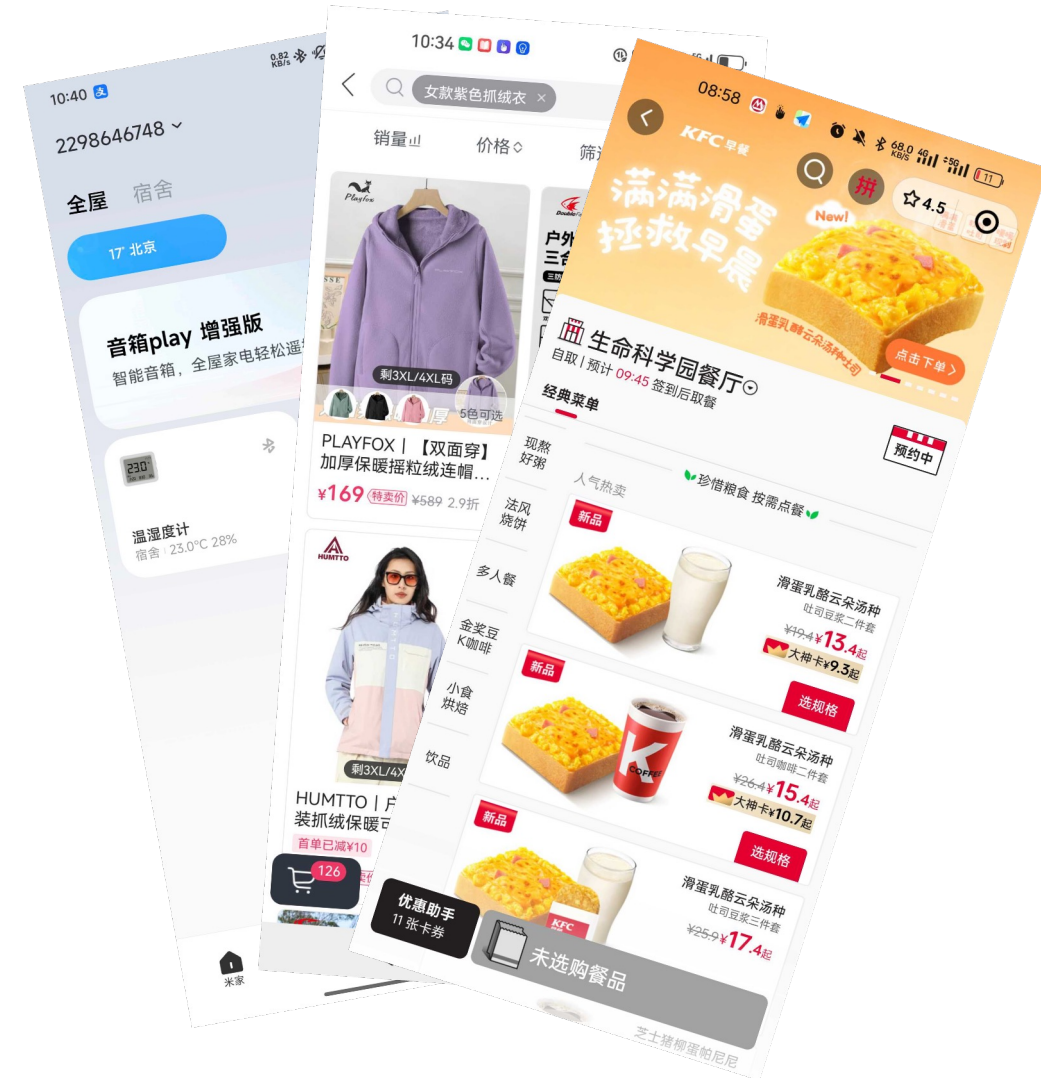
- Definition: the task that enables an AI to **see** Graphical User Interfaces (GUIs), **understand** the function of visual elements, and **link** them to abstract language commands.
 - Moves beyond simple object detection to functional understanding
 - Mapping abstract instructions (e.g. booking hair salon) to UI elements.
- Core Technologies: A fusion of Computer Vision (Object Detection, OCR) and Multimodal Language Model
- Goal: Create AI that can **control** any application just like us.



Limitation of Current Models

Problem Definition

- The app universe is **infinite**.
 - In 2024, Google Play store has **1.6M+ apps**
 - Impossible to train a single agent for every apps in world
- “Personal Gap” in Generic Models
 - Current models are trained on common generic UIs
 - Fail at indie app, customizing, or corporate software
- Adapting to the Personalized Environment
 - Average korean user has only **~102 apps**
 - requires **fine-tuning** the agent on each individual user



Dataset Description

FingerTip 20K

Dataset Description

- **The only user-labeled mobile GUI interaction dataset**, collected from 91 real users across 506 apps over long-term daily app usage.
- Each sample includes user profile, intent, scenario, and action logs, making it ideal for personalized GUI grounding and action modeling.

FingerTip 20K: A Benchmark for Proactive and Personalized Mobile LLM Agents

Qinglong Yang

Haoming Li

Haotian Zhao

Xiaokai Yan

Jingtao Ding

Fengli Xu

Yong Li

Directory Structure

Dataset Description

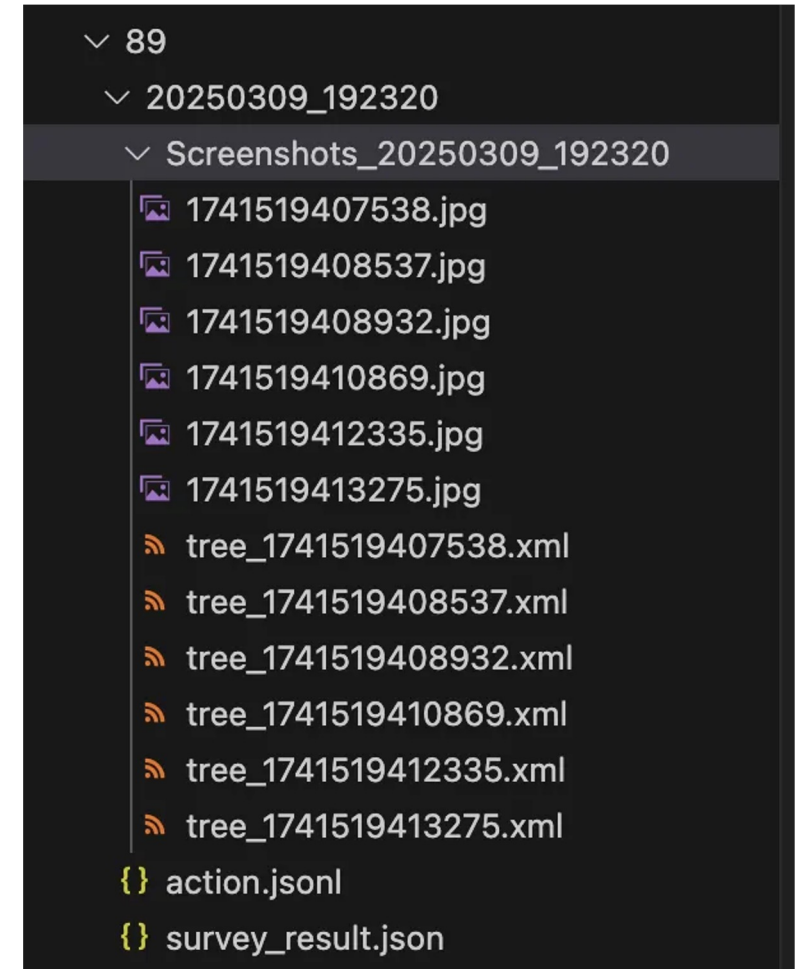
→ User id

→ Timestamp

→ Screenshots

→ Action

→ Survey__result



Screenshots_[TIMESTAMP]

Dataset Description



action.jsonl

Dataset Description – Code Structure

This file is a log file **recording a sequence of user actions** performed on the GUI

action	“click()”, “scroll()”, “press_back()”
coordinates	indicate the click or scroll position
content	describes the UI element or text at that location
direction	specifies the scroll direction (if applicable)

`{}` *action.jsonl* ×

dataset > 1 > 20250309_133115 > `{}` *action.jsonl*

```
1 {"1741498279844.jpg": "click(coordinates=(336, 1267), content='蓝色加号按钮')"}
2 {"1741498281851.jpg": "click(coordinates=(336, 1268), content='蓝色加号按钮')"}
3 {"1741498282279.jpg": "scroll(coordinates=(200, 630), direction='up')"}
4 {"1741498283792.jpg": "click(coordinates=(605, 142), content='√')"}
5 {"1741498287473.jpg": "click(coordinates=(340, 1189), content='+')"}
6 {"1741498288057.jpg": "finished()"}
7
```

survey_result.json

Dataset Description – Code Structure

This file stores metadata describing the user’s **GUI interaction scenario**

user_id	the identifier of the user participating in the data collection
time	the timestamp of the recorded session
scenario	the context or situation (e.g., school, home, office)
app	the package name of the application being used
intentDescription	a natural-language description of the user’s intended action or goal.

```
{ } survey_result.json ×
dataset > 1 > 20250309_133115 > { } survey_result.json > ...
1  {
2    "user_id": "1",
3    "time": "20250309_133115",
4    "scenario": "学校",
5    "app": "com.huawei.deskclock",
6    "intentDescription": "打开闹钟，设置一个二十分钟后的闹铃"
7  }
```

tree__[TIMESTAMP].xml

Dataset Description – Code Structure

This file provides **the View Hierarchy** corresponding to each screenshot in XML.

class	the type of UI element
text	the visible text or description
bound	The exact (x, y) coordinates of the element.
boolean properties	True/false flags, defining the current functional state of a UI element.

```
<node
  class="android.widget.FrameLayout"
  text=""
  content_desc=""
  view_id=""
  bounds="[0,0][1080,2340]"
  hint=""
  tooltip=""
  checked="false"
  enabled="true"
  focused="false"
  selected="false"
  clickable="false"
  long_clickable="false"
  checkable="false"
  focusable="false"
  editable="false"
  scrollable="false"
  password="false"
  visible="true">
```

Proposed Approach

Backbone Model

Proposed Approach

- We use **Qwen3-VL** as a fixed backbone for GUI understanding and grounding.
- This **state-of-the-art** multimodal model developed by Alibaba provides strong vision-language reasoning, OCR, and UI element comprehension.
- It offers excellent alignment between visual perception and instruction following,
- serving as a universal foundation while personalization is achieved through lightweight fine-tuning.



Fine-tuning Personalization

Proposed Approach

- Collect each user's real app data (screenshots, command logs, intents).
- Apply **fine-tuning method** (e.g. LoRA, PEFT) to update only a few user-specific parameters.
- The model adapts to the and unique interface patterns — achieving personalization without retraining the full model.



Thank You

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